Open Project QCG, IITR

Quantum Random Walks (in Bond Pricing)

Assignment 2

Note: All the questions this week are to be typed and submitted as a PDF file. Submission link will be sent soon.

Question 1

You have purchased a 10-year corporate bond that pays a handsome 7% annual coupon. However, the issuer has embedded a defined rate trigger: if the prevailing central bank interest rate (currently 5%) ever drops to exactly 3%, the bond is automatically called (canceled). The issuer instantly returns your principal, and you receive no further coupon payments.

Modelling this bond requires connecting financial mechanics to a concept called a "random walk with a barrier." Answer the following questions in order to clear your concepts on the same.

Part 1: Bond Financial Mechanics

Assume for a moment that interest rates move in a perfectly straight, predictable line.

1. Why did the corporate issuer include this 3% trigger in the bond's contract? What is their financial incentive?
2. From the investor's perspective, describe the cash flow of this bond as walking on a solid floor. What does the 3% trigger represent in this analogy, and what happens to the investor's expected cash flows the moment they step on it?

Part 2: Stochastic Movement

In reality, interest rates rarely move in straight lines. They react daily to inflation data, geopolitical news, and market panic. They take a "random walk", bumping up by a few basis points one day, dropping the next, wandering unpredictably.

1. If the current rate is 5%, and the rate takes a random walk every single day for 10 years, why is the bond *more* dangerous to the investor than if the rate just stayed perfectly still at 5%?
2. How does the concept of "volatility" (the size of the daily up-and-down steps in the random walk) change the likelihood of hitting the 3% mark?

Part 3: Model Synthesis

In quantitative finance, this 3% rate trigger is mathematically defined as a "down-and-out barrier." The moment the random walk touches this boundary line, the simulation stops.

Imagine running 1,000 different simulated paths (random walks) for the interest rate over the next 10 years. In 400 of those paths, the line dips and touches 3% at some point. In 600 paths, it wanders around but stays above 3% for the full 10 years.

1. How do we use the results of those 1,000 random walks to determine the *fair price* of this bond today? Explain conceptually how the "barrier hits" actually drag down the mathematical value of the bond.

Question 2

We're looking at exotic bond features called barriers (specifically knock-in and knock-out structures) and comparing them to the Black-Cox structural credit model, which is a famous way of calculating a company's probability of going bankrupt.

Part 1: Barrier Mechanics

Imagine two different 5-year bonds linked to a stock index.

Bond A has a "Down-and-Out" barrier at 80: If the index drops to 80, the bond is instantly canceled, paying zero future coupons.

Bond B has a "Down-and-In" barrier at 80: The bond pays no coupons *unless* the index drops to 80. If it hits 80, the bond "activates" and starts paying a massive 15% coupon.

1. In both bonds, what is taking the random walk?
2. Assume a standard, risk-free bond with no barriers costs $100. Conceptually, why must Bond A (Down-and-Out) cost *less* than $100? Why does higher volatility in the stock index make Bond A even cheaper?

Part 2: Defining the Walker and the Wall

In 1976, Fischer Black and John Cox proposed a structural model for pricing corporate debt. They argued that a corporation defaults exactly when the total value of its assets falls below a critical threshold (its liabilities/debt).

1. In the Black-Cox model, what specific financial variable is taking the "random walk"? What real-world metric represents the "barrier"?
2. In a simple rate-trigger bond, the barrier is a flat horizontal line (e.g., exactly 3%). But in the Black-Cox model, the default barrier often *slopes upward* over time. From a corporate finance perspective, why would a company's "default

threshold" naturally rise as the years go by? *(Hint: Think about what happens to unpaid debt or promised interest over time).*

Part 3: The Grand Synthesis

A standard corporate bond pays regular coupons and returns principal at maturity, *provided the company does not go bankrupt*.

1. Explain how buying a standard 10-year corporate bond from Ford Motor Company is mathematically identical to buying a 10-year risk-free US Treasury bond that has a "Down-and-Out" barrier attached to it.
2. n this analogy, what exactly is the "knock-out" event? What happens to your cash flows when the barrier is breached, and how does the Black-Cox model determine if that breach occurs?
3. If the market suddenly decides that Ford's business model is highly uncertain (meaning the random walk of their asset value takes wild, violent swings every day), what happens to the probability of hitting the barrier? Consequently, what must happen to the price of Ford's bond today?